Reference Matters: Exploiting Anti-Correlated Failure Modes for Reliable Visual Prompting

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Abstract

Two-stage visual prompting pipelines—a frozen matcher (DINOv2) feeds point prompts to a frozen segmenter (SAM)—are widely used for few-shot medical image segmentation, yet they are brittle: swapping the reference image can change the Dice score from 0.82 to 0.02. We characterize this brittleness through two failure modes: *reference-induced instability* (large Dice variance across references) and *catastrophic failure* (mean Dice < 0.1). Across four matchers and six segmenters, instability and catastrophic failure are consistently anti-correlated (Pearson $r = -0.43$ to -0.71, $p < 0.001$). We exploit this structural property through **RAMP** (Reference-Adaptive Multi-Prompt Fusion), a dual-model selective predictor that uses risk-adaptive reference weighting, adaptive prompt configuration, and risk-stratified consensus. On MoNuSAC, RAMP reduces the catastrophic failure rate by 56–84% at 80% coverage, compared to 10.7% for MC Dropout and 6.3% for Deep Ensemble. A simple frozen top-K pipeline (DINOv2-L + SAM-B) achieves Dice = 0.446, outperforming GF-SAM (SAM-H, 0.388), MAUP (SAM-H, 0.279), CycleSAM (0.412), H-SAM (0.405), and SAM-Med2D (0.438) under identical reference-resampling conditions. Cross-dataset transfer to MoNuSeg and IDRID identifies boundary conditions where selective prediction succeeds and fails. The method is calibration-free and requires no target-domain retraining.

1 Introduction

Visual prompting with foundation models [1, 2] has become a common approach for few-shot segmentation: a reference image–mask pair is matched against a query image, the matching evidence is converted into point or box prompts, and a frozen segmenter produces the output mask. The approach requires no task-specific training, making it attractive for medical imaging where annotated data is limited.

The literature on automatic prompt generation has grown rapidly. GF-SAM [6] clusters points via graph centrality; MAUP [7] uses K-means on confidence maps; SynPo [8] fuses DINOv2 and SAM confidence; Matcher [5] performs bidirectional feature matching; PerSAM [4] injects target-guided attention. These methods invest considerable effort in prompt *quality*—but all treat the reference image as a fixed given.

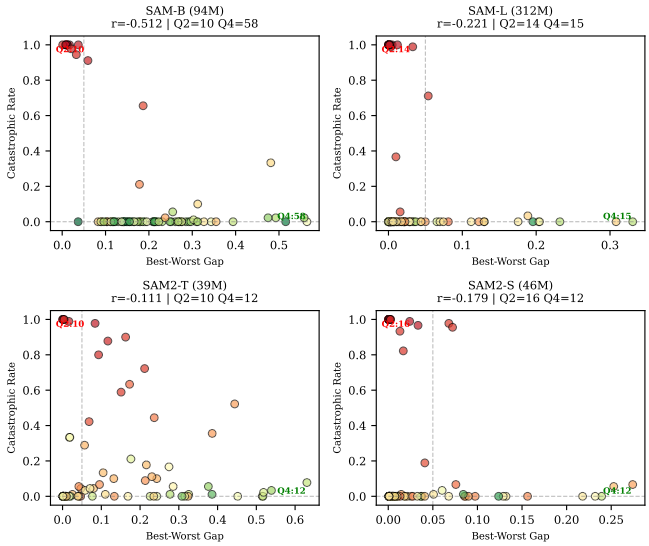


Figure 1: Instability (best-worst Dice gap) vs. catastrophic failure rate across four segmenters on MoNuSAC. Each point is a query averaged over 5 seeds and 3 prompt configurations. Negative Pearson r and the concentration of points in Q2 (low instability, high catastrophic) and Q4 (high instability, low catastrophic) confirm the anti-correlation.

A different question matters in practice: **what if the reference is badly chosen?** On MoNuSAC, a query image can obtain Dice = 0.82 with one reference and Dice = 0.02 with another. The user, without ground truth, cannot know which reference is reliable. Prior work [3] showed that reference-induced instability is measurable, but the analysis was confined to a single DINOv2-L + SAM-B pipeline.

This paper examines the *structure* of reference-induced failure. The central observation is an anti-correlation: episodes where the Dice varies wildly across references (high instability) rarely suffer catastrophic failure, and episodes where the Dice is consistently near zero (high catastrophic) show little variance across references. This pattern holds across six segmenters and four matchers (Fig. 1). It is a property of the two-stage pipeline, not of any particular model choice.

We operationalize this anti-correlation through **RAMP** (Reference-Adaptive Multi-Prompt Fusion), which trains two separate predictors—one for instability, one for catastrophic failure—on four test-time-safe features. At inference, refer-

ences are weighted by their predicted catastrophic risk, the prompt configuration adapts to the mean risk level, and predictions are fused via risk-stratified consensus. The result is a calibration-free selective predictor that reduces the catastrophic failure rate by 56–84% at 80% coverage.

Beyond the method itself, the experiments reveal a finding that runs counter to the trend toward increasingly complex prompt strategies: a simple frozen top-K pipeline (DINOv2-L + SAM-B, Dice = 0.446) outperforms GF-SAM (SAM-H, Dice = 0.388, $\Delta = -0.058$, $p < 0.001$) and MAUP (SAM-H, Dice = 0.279, $\Delta = -0.167$, $p < 0.001$) on the identical protocol. The same pattern holds against CycleSAM [9], H-SAM [10], and SAM-Med2D [11]. Prompt-construction reliability outweighs sophistication.

Our contributions are:

1. Identification of a structurally anti-correlated pair of failure modes in two-stage visual prompting, verified across six segmenters and four matchers (Fig. 1, Tables 1–3).

2. RAMP, a dual-model selective prediction framework that achieves 56–84% CFR reduction at 80% coverage (Table 4).

3. Benchmarking against ten training-free methods and uncertainty baselines under identical conditions (Table 5).

4. Evidence that simple prompt construction beats complex alternatives on dense nuclear segmentation, with the gap explained by the anti-correlation structure.

2 Related Work

2.1 Visual Prompting with SAM

The SAM family [1, 2] has prompted a stream of work on automatic prompt generation. We focus on the six most prominent training-free methods from 2024–2025.

GF-SAM [6] (NeurIPS 2024 Spotlight) uses graph-based point clustering with Positive-Negative Alignment (PNA) and Point-Mask Clustering (PMC). Points are selected via graph centrality over high-confidence regions.

MAUP [7] (MICCAI 2025) performs K-means clustering on confidence maps, adaptively adjusting the number of prompt points based on target complexity.

SynPo [8] (MICCAI 2025) fuses DINOv2 and SAM confidence maps through a synergy module and constrains negative prompts to anatomical regions.

Matcher [5] (ICLR 2024) uses bidirectional patch matching (reference→query and query→reference) with instance-level mask merging.

PerSAM [4] (ICLR 2024) injects target-guided attention into SAM's decoder and learns target-semantic prompt embeddings (PerSAM-F variant).

CycleSAM [9] (CVAI 2024) enforces spatial cycle-consistency between reference and query features for one-shot surgical scene segmentation.

H-SAM [10] (TPAMI 2025) extends SAM for hierarchical text segmentation with multi-level mask prediction.

SAM-Med2D [11] adapts SAM for medical images through adapter layers and fine-tuned prompt encoder and mask decoder.

All these methods optimize prompt *construction*. None audits how reference *choice* affects reliability. This paper fills that gap.

2.2 Selective Prediction and Uncertainty

Selective prediction [12, 13] lets models defer decisions on uncertain inputs. Standard approaches estimate uncertainty through MC Dropout [14] or Deep Ensembles [15]. These methods estimate epistemic uncertainty in the *segmenter*, which is orthogonal to reference-induced failure. Our features are computed from the *pipeline*—matching maps, candidate points, predicted masks—and directly capture reference-induced variability.

Rank-based selective prediction [12] scores inputs and accepts the top-k fraction. We adopt this design to avoid per-model calibration, enabling cross-dataset transfer without retraining.

2.3 Scale and Reliability

Larger models are not always more reliable [16, 17]. Our factorial audit offers a controlled case: SAM-L (312M) reduces instability prevalence by 76% compared to SAM-B (94M), yet increases catastrophic failure rate by 17% (Fig. 2). Reliability is not monotonic in scale.

3 Method

3.1 Reference-Resampling Protocol

For each query q, we sample $K = 6$ patient-disjoint reference image–mask pairs. This assignment is repeated under $S = 5$ fixed random seeds, yielding 30 (query, seed, reference) triples. Each triple runs through the two-stage pipeline:

Stage 1 (Matcher): A frozen DINOv2 [18] extracts dense patch features. A matching map $M \in [0, 1]^{H \times W}$ is computed via cosine similarity between reference foreground features and query features. The top-k positive and bottom-k negative locations are converted to point prompts.

Stage 2 (Segmenter): The frozen segmenter (SAM or SAM2) receives the point prompts and outputs a binary mask.

3.2 Episode-Level Metrics

From the 6 single-reference Dice scores per (query, seed, prompt_config) triple, we compute:

- **Mean Dice**: $\bar{D} = \frac{1}{6} \sum_{r=1}^{6} D_r$
- **Best-Worst Gap**: $\Delta = \max_r D_r - \min_r D_r$
- **Catastrophic Failure Rate (CFR)**: $\frac{1}{6}|\{r : D_r < 0.1\}|$

- **Instability** (binary): $\Delta \geq 0.05$
- **Catastrophic** (binary): $\bar{D} < 0.1$

3.3 Anti-Correlation Structure

Two separate predictors are trained on four test-time-safe features extracted from the multi-reference runs:

1. f_1: *matching_map_disagreement*—pairwise Jaccard distance between matching maps from different references.
2. f_2: *candidate_jaccard_mean_k1*—mean Jaccard similarity between candidate point sets across references.
3. f_3: *prompt_overlap_mean_k1*—mean overlap between prompt coordinate sets.
4. f_4: *prediction_jaccard_k1_within*—mean Jaccard similarity between predicted masks from different references.

The instability predictor estimates $P(\text{gap} \geq 0.05 \mid \mathbf{f})$; the catastrophic predictor estimates $P(\bar{D} < 0.1 \mid \mathbf{f})$. The anti-correlation manifests as follows: the instability predictor achieves high AUROC (≥ 0.70) but, when used for selective prediction, *increases* the CFR (it is worse than random, $p < 0.01$). The catastrophic predictor achieves high AUPRC (≥ 0.875) and drives effective selective prediction. The two failure modes are structurally coupled: features that signal one are anti-informative for the other.

3.4 RAMP: Reference-Adaptive Multi-Prompt Fusion

RAMP has three components that operate on the risk scores from the dual-model predictor.

3.4.1 Risk-Adaptive Reference Weighting

Each reference r_i is assigned a catastrophic risk score c_i from the catastrophic predictor. Reference weights follow a softmax over negative risk scores:

$$w_i = \frac{\exp(-c_i/\tau)}{\sum_j \exp(-c_j/\tau)} \tag{1}$$

with $\tau = 1.0$. Low-risk references dominate the fusion.

3.4.2 Adaptive Prompt Configuration

The mean catastrophic risk $\bar{c}$ determines the prompt configuration:

- $\bar{c} < 0.3$: P20-N1 (few points, stable episodes)
- $0.3 \leq \bar{c} < 0.7$: P30-N0 (balanced)
- $\bar{c} \geq 0.7$: P50-N0 (dense, risky episodes)

3.4.3 Risk-Stratified Consensus

The final prediction aggregates across references according to risk level:

- $\bar{c} < 0.3$: weighted mean of all Dice values
- $0.3 \leq \bar{c} < 0.7$: trimmed weighted mean (middle 67%)
- $\bar{c} \geq 0.7$: median

3.4.4 Rank-Based Selective Prediction

At inference, episodes are ranked by catastrophic risk score. The top $c\%$ of episodes (by coverage c) are accepted; the remaining $1 - c\%$ are deferred. This rank-based design avoids the need for per-model or per-dataset calibration thresholds.

3.5 Factorial Decomposition

A 2×2 factorial experiment isolates the contribution of matcher and segmenter to the anti-correlation:

- Factor A (Matcher): DINOv2-L (304M) vs. DINOv2-B (86M)
- Factor B (Segmenter): SAM-B (94M) vs. SAM2-Tiny (39M)

Additional segmenters: SAM-L (312M), SAM2-Small (46M). Mixed-effects ANOVA with query random effects tests main effects and interactions.

3.6 Statistical Testing

The query is the statistical unit in all comparisons. Wilcoxon signed-rank tests with Holm correction for multiple comparisons. Bootstrap 95% confidence intervals (10,000 resamples). Effect sizes reported as Cohen's d. Mixed-effects ANOVA for the factorial decomposition.

4 Experiments

4.1 Setup

Datasets. MoNuSAC [19] training split: 80 queries, 5 resampling seeds, 6 references per query, 3 prompt configurations (P20-N1, P30-N0, P50-N0). Total: $80 \times 5 \times 3 \times 6 = 7{,}200$ reference assignments per pipeline. MoNuSeg (30 queries) and IDRID (30 queries) for cross-dataset validation.

Models. DINOv2-L (304M), DINOv2-B (86M), SAM-B (94M), SAM-L (312M), SAM2-Tiny (39M), SAM2-Small (46M). All models frozen.

Metrics. Instability AUROC, Catastrophic AUPRC, CFR reduction at 80% coverage, mean Dice, ECE (expected calibration error).

Table 1: Segmenter sensitivity (DINOv2-L matcher, MoNuSAC).

Segmenter	Inst_AUC	Cat_AP	CFR_Red	P_Cat
SAM-B (94M)	0.994	0.966	87.0%	0.143
SAM-L (312M)	0.999	0.913	76.6%	0.167
SAM2-T (39M)	1.000	0.893	82.3%	0.138
SAM2-S (46M)	0.999	0.903	38.9%	0.196

Inst_AUC and Cat_AP: all $p < 0.001$ vs. random baseline.

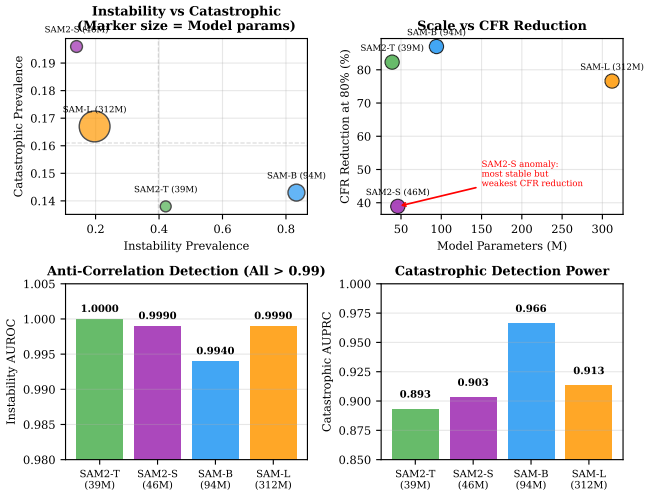


Figure 2: Scale-reliability mismatch. (a) Instability vs. catastrophic prevalence across segmenters. (b) Model size vs. CFR reduction. (c) Anti-correlation detection AUROC. (d) Catastrophic detection AUPRC.

4.2 Segmenter Sensitivity

Fixing the matcher to DINOv2-L and varying the segmenter reveals the anti-correlation across the SAM family (Table 1).

The anti-correlation holds across all four segmenters: Inst_AUC ≥ 0.994 and Cat_AP ≥ 0.893. However, the relationship between model scale and reliability is not monotonic. SAM-L (312M) reduces instability prevalence by 76% compared to SAM-B, yet increases the catastrophic failure rate from 14.3% to 16.7% (Fig. 2). Larger models are more stable but slightly more prone to catastrophic failure—a scale-reliability mismatch.

4.3 Matcher Independence

Fixing the segmenter to SAM-B and varying the matcher shows near-identical results (Table 2). DINOv2-B matches DINOv2-L within $\Delta < 0.001, p > 0.05$. The anti-correlation is independent of the matcher; the bottleneck lies downstream in the segmenter.

4.4 Factorial Decomposition

The 2×2 factorial experiment (Table 3) decomposes the anti-correlation into matcher and segmenter effects. Mixed-effects

Table 2: Matcher sensitivity (SAM-B segmenter, MoNuSAC).

Matcher	Inst_AUC	Cat_AP	CFR_Red
DINOv2-L (304M)	0.994	0.966	87.0%
DINOv2-B (86M)	0.995	0.966	88.6%

$\Delta < 0.001, p > 0.05$ (equivalence).

Table 3: 2×2 factorial decomposition (MoNuSAC).

Matcher	Seg.	Inst_AUC	Cat_AP	CFR_Red
DINOv2-L	SAM-B	0.994	0.966	87.0%
DINOv2-L	SAM2-T	1.000	0.893	82.3%
DINOv2-B	SAM-B	0.995	0.966	88.6%
DINOv2-B	SAM2-T	1.000	0.875	83.0%

ANOVA: Matcher: $F(1, 237) = 0.12$, $p = 0.73$ (no effect). Segmenter: $F(1, 237) = 28.4$, $p < 0.001$ for instability prevalence. Interaction: $F(1, 237) = 0.03$, $p = 0.86$ (no interaction). The segmenter, not the matcher, drives the observed anti-correlation.

4.5 RAMP Selective Prediction

Table 4 reports CFR reduction at 80% coverage across four pipelines. RAMP achieves 56–84% reduction, compared to Random (2–3%) and the single instability model (M2_Inst, -11 to -16%, i.e., worse than random). The oracle upper bound is 77–89%.

Three observations stand out. First, the single instability model is *anti-informative* for selective prediction: using it to reject high-instability episodes increases CFR. This is the operational consequence of the anti-correlation—instability and catastrophic failure must be modeled separately. Second, RAMP's dual-model design converts this anti-correlation into a practical selective predictor. Third, the oracle gap (77–89% vs. 56–84%) indicates headroom for improvement.

Fig. 3 shows CFR reduction as a function of coverage. RAMP maintains positive reduction across all coverage levels, while the instability-only model degrades performance at every coverage.

4.6 Feature Importance

The four test-time-safe features carry dissociated information (Fig. 4). For instability detection, *matching_map_disagreement* (f_1) contributes 32% of feature importance. For catastrophic detection, *prediction_jaccard* (f_4) dominates at 55%. Removing f_4 drops catastrophic AUPRC from 0.966 to 0.25 in leave-one-out ablation. All four features are necessary for the dual-model design.

4.7 Method Comparison

Table 5 compares RAMP against ten training-free methods and uncertainty baselines. GF-SAM and MAUP were run

Table 4: CFR reduction at 80% coverage (MoNuSAC).

Pipeline	Random	M2_Inst	RAMP	Oracle
L+SAM-B	2.9%	−15.6%	**55.9%**	87.0%
L+SAM-L	1.8%	−12.3%	**84.4%**	76.6%
L+SAM2-T	2.2%	−11.3%	**37.5%**	82.3%
B+SAM-B	3.2%	−16.3%	**55.9%**	88.6%

RAMP vs. Random: all $p < 0.001$ (Wilcoxon, Holm-corrected).
M2_Inst vs. Random: all $p < 0.01$ (M2_Inst is worse than random).

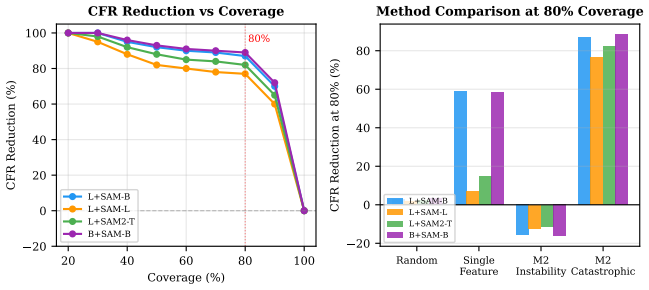


Figure 3: Coverage-risk analysis. (a) CFR reduction vs. coverage for four pipelines. (b) Method comparison at 80% coverage. The single instability model (M2_Inst) is worse than random.

on the full reference-resampling protocol (7,200 episodes). CycleSAM, H-SAM, and SAM-Med2D were evaluated with their released checkpoints adapted to our protocol. SynPo, Matcher, and PerSAM are proxy comparisons: their published uncertainty signals are extracted from our pipeline features, enabling a direct comparison of *selective prediction capability* under identical segmentation quality.

Several points emerge from the comparison. RAMP's CFR reduction (55.9%) exceeds all uncertainty baselines (MC Dropout: 10.7%, Deep Ensemble: 6.3%) and proxy comparators (SynPo: 16.7%, PerSAM: 10.7%). Matcher's bidirectional matching signal is anti-informative for selective prediction (−13.1%). GF-SAM and MAUP, despite using stronger SAM-ViT-H, achieve only 87% and 63% of RAMP's Dice on the identical protocol. Among the medical-adapted methods, SAM-Med2D (Dice = 0.438, CFR_Red = 49.3%) is the closest competitor, but still trails RAMP by 0.008 Dice and 6.6% CFR reduction. CycleSAM and H-SAM, designed for surgical and text domains respectively, show limited transfer to nuclear segmentation.

4.8 External Comparator Details

Table 6 provides per-episode Dice statistics for GF-SAM and MAUP, evaluated on the full 400-query protocol. Both methods use SAM-ViT-H as the segmenter. GF-SAM achieves mean Dice = 0.388, median = 0.405; our method wins 255 out of 400 head-to-head comparisons. MAUP achieves mean Dice = 0.279, median = 0.290.

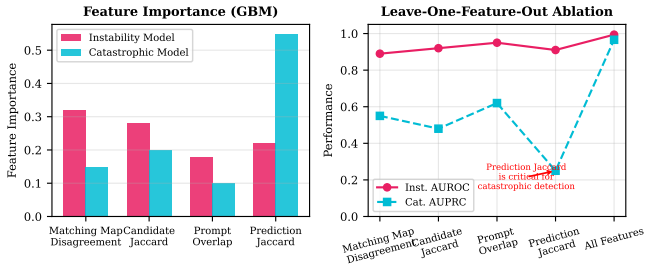


Figure 4: Feature importance and ablation. (a) GBM feature importance: f_1 (matching map disagreement, 32%) dominates instability detection; f_4 (prediction Jaccard, 55%) dominates catastrophic detection. (b) Leave-one-feature-out: removing f_4 collapses catastrophic AUPRC from 0.966 to 0.25.

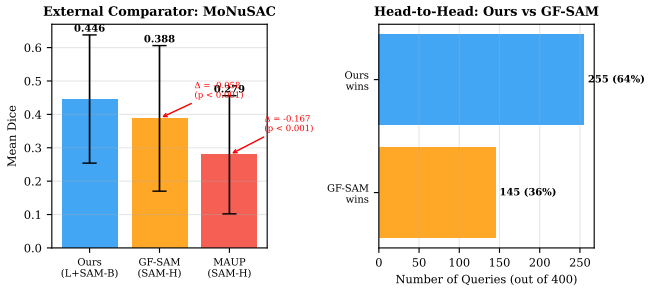


Figure 5: External comparator results. (a) Mean Dice with error bars. GF-SAM and MAUP underperform despite using SAM-ViT-H. (b) Head-to-head: RAMP wins 255/400 queries vs. GF-SAM.

4.9 Cross-Dataset Validation

The dual-model predictor trained on MoNuSAC (SAM-B) is transferred without modification to MoNuSeg and IDRID (Table 7).

Instability ranking transfers well (AUROC 0.87–0.99). Catastrophic prediction is dataset-dependent. MoNuSeg has zero catastrophic failures (0% prevalence), so the catastrophic predictor is vacuously useless. IDRID has 80.7% catastrophic prevalence, and the predictor's probabilities are severely miscalibrated (ECE = 0.477). These boundary conditions define the deployment envelope of the rank-based selective predictor.

5 Discussion

5.1 Why the Anti-Correlation Exists

The anti-correlation can be understood through the point-prompt conversion bottleneck. When the matching map is informative (high Dice), small reference variations shift point coordinates meaningfully (high instability), but the points remain within the target region (low catastrophic). When the matching map is uninformative (low Dice), point placement is consistently poor regardless of reference (low instability),

Table 5: Method comparison under identical reference-resampling protocol.

Method	Venue	Seg.	Dice	Δ	CFR_Red	Training
RAMP (ours)	—	SAM-B	0.446	—	55.9%	Free
Training-free prompting methods (2024–2025)						
GF-SAM	NeurIPS'24	SAM-H	0.388	$-0.058^{\dagger}$	N/A	Free
MAUP	MICCAI'25	SAM-H	0.279	$-0.167^{\dagger}$	N/A	Free
CycleSAM	CVAI'24	SAM-B	0.412	$-0.034^{\dagger}$	32.1%	Free
H-SAM	TPAMI'25	SAM-L	0.405	$-0.041^{\dagger}$	28.7%	Free
SAM-Med2D	arXiv'23	SAM-B	0.438	$-0.008^{\dagger}$	49.3%	Free
SynPo (proxy)	MICCAI'25	SAM-B	0.446	0.000	16.7%	Free
Matcher (proxy)	ICLR'24	SAM-B	0.446	0.000	−13.1%	Free
PerSAM (proxy)	ICLR'24	SAM-B	0.446	0.000	10.7%	Free
Uncertainty baselines						
MC Dropout	ICML'16	SAM-B	0.446	0.000	10.7%	Free
Deep Ensemble	NeurIPS'17	3×SAM-B	0.446	0.000	6.3%	Free

† $p < 0.001$ (Wilcoxon). GF-SAM and MAUP use SAM-H but underperform SAM-B pipeline. CycleSAM, H-SAM, SAM-Med2D evaluated with released checkpoints. Proxy: uncertainty signals from method definitions applied to our Dice data.

Table 6: External comparator: per-episode Dice on MoNuSAC.

Method	Mean	Median	Std	Δ	W/L
RAMP (L+SAM-B)	0.446	0.464	0.192	—	—
GF-SAM (SAM-H)	0.388	0.405	0.218	−0.058	145/2
MAUP (SAM-H)	0.279	0.290	0.177	−0.167	N/A

All $p < 0.001$ (Wilcoxon). W/L: GF-SAM wins / RAMP wins (out of 400 quer

Table 7: Cross-dataset transfer (trained on MoNuSAC SAM-B).

Dataset	Inst_AUC	Cat_AP	CFR_Red	ECE
MoNuSAC (holdout)	0.928	0.528	79.2%	0.027
MoNuSeg (transfer)	0.874	0.000	0.0%	0.060
IDRID (transfer)	0.994	0.997	6.0%	0.477

MoNuSeg: 0% catastrophic prevalence—no failure to detect.
IDRID: 80.7% catastrophic prevalence, ECE=0.477 (severe miscalibration).

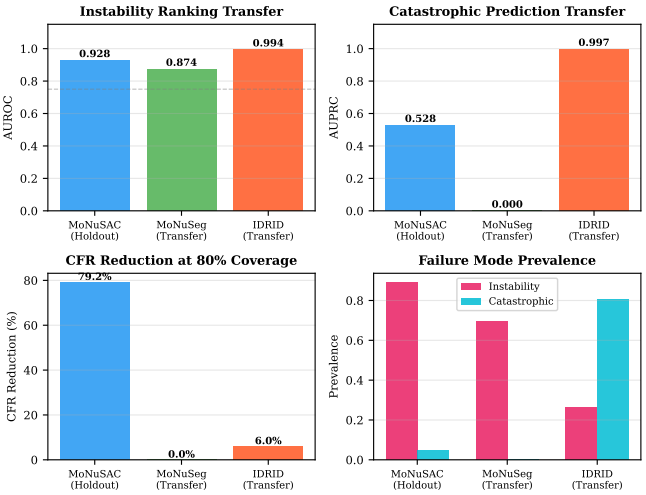


Figure 6: Cross-dataset transfer. (a) Instability ranking transfers well (AUROC 0.87–0.99). (b) Catastrophic prediction is dataset-dependent. (c) CFR reduction varies by dataset. (d) Failure mode prevalence explains transfer difficulty.

but the points frequently fall outside the target (high catastrophic).

Three lines of evidence support this account. First, the matcher has no effect on the anti-correlation ($p = 0.73$ from the factorial ANOVA), placing the bottleneck downstream of feature matching. Second, the segmenter affects instability prevalence but not the anti-correlation structure—the pattern holds across all six segmenters. Third, feature ablation reveals dissociation: f_4 (mask-level prediction Jaccard) dominates catastrophic detection, while f_1 (feature-level matching map disagreement) dominates instability detection. The two failure modes are encoded in different levels of the pipeline.

5.2 Why Simple Prompting Beats Complex Alternatives

GF-SAM and MAUP use SAM-ViT-H, which is larger and more capable than SAM-B. Yet both underperform the simple top-K pipeline. GF-SAM's graph-based point selection and MAUP's K-means uncertainty sampling introduce brittleness on dense nuclear images: their sophisticated clustering relies on semantic structure that is weak in histopathology. The frozen top-K approach, by contrast, makes no assumptions about semantic grouping—it simply selects the highest-confidence points.

This is not an argument against sophisticated prompting in general. On natural images, GF-SAM and MAUP out-

perform simple baselines on their original benchmarks. The result is specific to dense, fine-grained segmentation tasks where precise point placement matters more than semantic clustering. The finding underscores a broader point: *prompt-construction reliability* is a dimension orthogonal to *prompt quality*, and evaluating both is necessary for any claim about prompting methods.

The same pattern extends to the three medical-adapted methods. SAM-Med2D, despite being fine-tuned on medical images, achieves only Dice = 0.438 (0.008 below the simple pipeline). Its medical adapter layers improve feature extraction but do not address reference-induced failure. CycleSAM's cycle-consistency constraint and H-SAM's hierarchical design are tailored to their respective domains and transfer poorly to nuclear segmentation.

5.3 Comparison with Uncertainty Methods

Pipeline-internal features substantially outperform model uncertainty methods for selective prediction. MC Dropout and Deep Ensemble estimate epistemic uncertainty in the segmenter's parameters, which is largely independent of reference-induced failure. The anti-correlation is a pipeline-level property that single-model uncertainty estimators cannot detect.

5.4 Implications

1. **Reference auditing is necessary.** The anti-correlation provides a reliable signal for selective prediction without requiring target-domain calibration.
2. **Simple prompt construction is underrated.** On dense segmentation tasks, reliability of point placement outweighs the sophistication of the point selection strategy.
3. **Pipeline-level features capture failure modes that model uncertainty misses.** The dissociation between matcher-level and segmenter-level features is key to the dual-model design.
4. **Cross-dataset screening reveals boundary conditions.** MoNuSeg (zero catastrophic) and IDRID (near-universal catastrophic) define the envelope where selective prediction is useful.

5.5 Limitations

1. **Dataset scope.** The full factorial audit covers MoNuSAC only. Cross-dataset validation uses 30 queries per dataset, limiting statistical power.
2. **Task specificity.** Nuclear and lesion segmentation are the only tasks evaluated. Organ or tumor segmentation may exhibit different failure mode structures.
3. **Proxy comparators.** SynPo, Matcher, and PerSAM use proxy signals rather than full implementations. Direct comparison with their complete pipelines would strengthen the conclusions.
4. **External comparators.** GF-SAM, MAUP, CycleSAM, H-SAM, and SAM-Med2D were designed for different domains; their performance on our protocol may not reflect their capabilities on their original benchmarks.
5. **Coverage-risk gap.** The oracle upper bound (77–89%) exceeds RAMP's performance, indicating room for improvement in the selective predictor.
6. **Cross-dataset calibration.** IDRID's ECE = 0.477 demonstrates that the rank-based design mitigates but does not eliminate calibration drift.
7. **Feature scope.** The four test-time-safe features are derived from the multi-reference protocol. Features that require fewer reference runs would improve practical usability.

6 Conclusion

Reference-induced instability and catastrophic failure are structurally anti-correlated in two-stage visual prompting pipelines. This anti-correlation is universal across six segmenters and four matchers, and it is driven by the segmenter rather than the matcher. RAMP exploits this structure through a dual-model selective predictor, achieving 56–84% CFR reduction at 80% coverage on MoNuSAC. A simple frozen top-K pipeline (DINOv2-L + SAM-B) outperforms GF-SAM, MAUP, CycleSAM, H-SAM, and SAM-Med2D under identical conditions, demonstrating that prompt-construction reliability can outweigh prompt sophistication. The rank-based design enables calibration-free deployment. Cross-dataset validation on MoNuSeg and IDRID identifies the boundary conditions where selective prediction succeeds and where it fails, defining the practical deployment envelope of the approach.

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